

a The libraries every headline number runs on

Library	Species / accession	Chemistry	Upstream pipeline	Cells	Reads / input	R2 length (--seq-len)	Poly(A) clip rate (% of CB reads)
PBMC 10k v3 (human donor 1)	human 10x public pbmc_10k_v3	10x 3' v3	CellRanger 3.0.0 (GRCh38-3.0.0 refdata)	~10,000 (est.)	557,564,408 accepted CB reads	91 bp	0.5730 genome-wide qualifying (3,195,067 / 557,564,408) head-sample estimator: 2.2565
pbmc4k (human donor 2)	human 10x public pbmc4k (flow cell H53GNBCXY)	10x 3' v2	CellRanger 2.1.0 (GRCh38-1.2.0 refdata)	4,340 (est.)	87,433 mean reads/cell; BAM 33.24 GB	98 bp	head-sample estimator: 3.6965 (genome-wide not measured)
testis mouse 1 (GSE104556)	mouse GEO GSE104556	10x 3' v2	STARsolo (version not recorded)	2,500 (both mice together)	—	98 bp	—
testis mouse 2 (GSE104556)	mouse GEO GSE104556	10x 3' v2	STARsolo (version not recorded)	(see mouse 1)	—	98 bp	—
Laughney LUAD cohort (17 libs, cohort-level row)	human GEO GSE123904 17 GSMs = 14 patients	10x 3' (v2 era; HiSeq 2500)	—	29,063 curated; 18,651 label- confirmed (64.2%)	~224 GB of BAM; 505,197 unified PAS	—	—

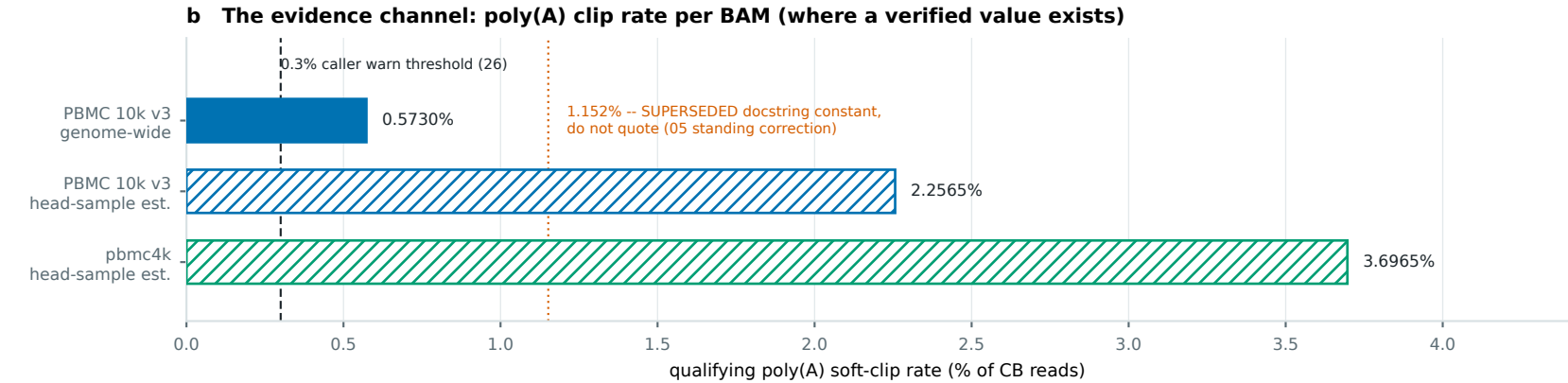


Fig S1 | The datasets behind every headline number, with the poly(A) clip rate -- the evidence channel the clip-seeded caller runs on. (a) Per-library table; all four benchmark arms ran the identical v2 parameter block (108 recorded parameters; the only library parameter is --seq-len = the R2 read length, read here from each arm's run_config.json). An em-dash means no verified measurement exists in the manuscript set and the value is omitted rather than approximated; testis clip rates (10 \$R2 required the measurement; none was recorded), testis/cohort per-BAM read counts, and the cohort clip rate -- the cohort's own 0.015% warning is invalid (head-of-chr1 sampling; 20) and is never repeated as fact. (b) Clip rates where a verified value exists. The genome-wide qualifying rate on PBMC 10k v3 is 0.5730% (3,195,067 / 557,564,408 accepted CB reads; 23 \$2) -- the corrected constant, superseding the 1.152% docstring figure. Hatched bars are the caller's head-sample QC estimator (first 200,000 CB reads of a coordinate-sorted BAM = head of chr1), which reads 2.2565% on PBMC where its own criteria measure 0.5364% genome-wide (~4.2x head bias; 23 \$3, 05 standing correction) -- so estimator values are comparable to each other, not to the genome-wide bar. pbmc4k's estimator value is 1.64x donor 1's: the evidence channel is richer on donor 2, and both are far above the 0.3% warn threshold (26 \$R4.4). Sources per cell: figS1_datasets.tsv.